

Speech Quality Measurement

440-2216/01 Quality of Service • Lecture 05

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Where we are going

Part 1

Subjective tests

Listeners, rating scales, MOS and its confidence interval

↔ the ground truth everything else predicts

Part 2

Signal-based models

PESQ, POLQA, ViSQOL — intrusive; P.563
— non-intrusive

↔ Lab 03 scores recordings with these

Part 3

The E-model

ITU-T G.107: from delay, loss and codec to a rating R

↔ Lectures 03 and 04 supply its inputs

One question runs through the lecture: **what does a quality number actually predict — and for which listeners, which task, and which audio bandwidth?**

What you should be able to do

1. Describe the ACR, DCR and CCR test methods of ITU-T P.800, compute a MOS with its confidence interval, and use the P.800.1 labels (MOS-LQS, -LQO, -CQE ...) correctly.
2. Distinguish intrusive, non-intrusive and parametric models, and outline how PESQ turns two signals into one score.
3. Choose a model that matches the audio bandwidth and the impairment being studied — and recognise when a model cannot see an impairment at all.
4. Compute the E-model rating R from delay, codec, packet loss and the advantage factor, and convert it to MOS and a user-satisfaction category.

Keep asking: **was this number obtained from people, from a signal, or from parameters — and was it a listening or a conversational situation?**

A quality score is a prediction of what listeners would say

Every objective method is only as good as the subjective test it was calibrated against

Part 1

Subjective assessment



Why ask people at all?

Lecture 03 separated **QoS** — what the network delivers — from **QoE** — what the user perceives. Speech quality is a QoE quantity: it exists only in the listener's judgement, so the **reference method is to ask listeners**.

Mean Opinion Score (MOS)

The mean of the values on a predefined scale that test subjects assign to their opinion of a system's performance (ITU-T P.10/G.100). A MOS is a property of a **condition** — a codec, a loss rate, a delay — averaged over many listeners and samples.

Methods of subjective determination — ITU-T P.800 (1996)

The standard defines how to run the test so that results from different laboratories can be compared: rating scales, listener selection, speech material, playback conditions and reference conditions.

Three rating scales of ITU-T P.800

ACR → MOS

one sample, no reference

5	Excellent
4	Good
3	Fair
2	Poor
1	Bad

DCR → DMOS

reference first, then degraded

5	Inaudible
4	Audible, not annoying
3	Slightly annoying
2	Annoying
1	Very annoying

CCR → CMOS

pair in random order

+3	Much better
+2	Better
+1	Slightly better
0	About the same
-1	Slightly worse
-2	Worse
-3	Much worse

MOS = arithmetic mean of all listeners' scores for one condition · always report it with its confidence interval

ACR is the default and the basis of almost every objective model. **DCR** is more sensitive to small degradations because the listener hears the clean reference first. **CCR** also detects *improvements* — for example a noise-suppression algorithm that makes the output better than its input.

How a listening test is run

Listeners

- **naive** — not involved in speech-quality work, so they rate as ordinary telephone users would
- tens of listeners per test, each hearing every condition

Material

- short sentence pairs (a few seconds) from several male and female talkers
- presented in **randomised order** to cancel learning and fatigue effects

Environment

- quiet, acoustically controlled room
- calibrated playback level through a standard handset or headphones

Anchors

- reference conditions such as the **MNRRU** (Modulated Noise Reference Unit, ITU-T P.810) — noise of known SNR added to the speech — let results of different laboratories be aligned

A single test with 24 listeners, 4 talkers and 20 conditions already needs almost 2 000 ratings — which is exactly why objective models exist.

MOS is a statistic — report it with its uncertainty

For one condition rated by N listeners with scores x_i :

$$\text{MOS} = \frac{1}{N} \sum_{i=1}^N x_i$$

$$s = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \text{MOS})^2}$$

$$\text{CI}_{95\%} = \text{MOS} \pm t_{0.975, N-1} \frac{s}{\sqrt{N}}$$

Three caveats (Streijl, Winkler & Hands, 2016):

- The ACR scale is **ordinal** — "Good" is not guaranteed to be exactly twice as far from "Poor" as "Fair" is. The mean is used by convention; distributions and percentages "Good or better" carry extra information.
- Listeners avoid the scale ends, so even clean speech rarely scores above ≈ 4.5 .
- MOS values depend on the **context of the test** — the same sample scores higher among bad conditions than among good ones.

Checkpoint — are the two codecs different?

In one ACR test with 24 listeners, codec A scores MOS 3.6 and codec B scores MOS 3.8. Both have a sample standard deviation of 0.8. Can you claim that codec B is better?

Predict first · discuss with a neighbour · then reveal

Not from these numbers. The 95 % half-width is $2.07 \cdot 0.8 / \sqrt{24} \approx 0.34$, so A lies in 3.6 ± 0.34 and B in 3.8 ± 0.34 — the intervals overlap widely. A difference of 0.2 MOS needs either many more listeners (the half-width shrinks only with $\sqrt{N}$) or a more sensitive method such as DCR or CCR, which compares the codecs directly.

Naming the result — P.800.1 labels, DMOS and CMOS

ACR results — ITU-T P.800.1 labels

Method	Listening	Conversational
Subjective	MOS-LQS	MOS-CQS
Objective	MOS-LQO	MOS-CQO
Estimated	MOS-LQE	MOS-CQE

Suffixes -N, -W, -S, -F: NB, WB, SWB, FB.

Listening — one recording: distortion, noise, loss. **Conversational** — two people talking: **delay and echo** count too. **O** — from signals (Part 2); **E** — from parameters (Part 3).

DCR and CCR results — ITU-T P.800

Method	Scale	Result
ACR	5 ... 1, absolute quality	MOS
DCR	5 ... 1, degradation vs. reference	DMOS
CCR	+3 ... -3, comparison of a pair	CMOS

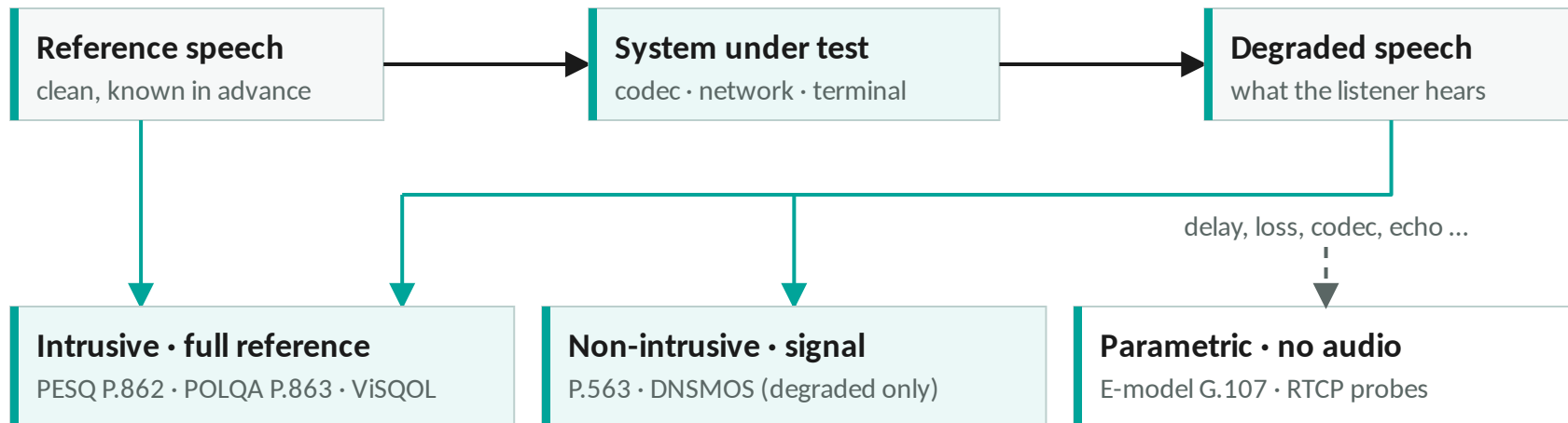
P.800.1 labels cover the **ACR scale only** — DMOS and CMOS keep their own names and are never mixed with MOS:

- **DMOS 4** = "audible but not annoying", not "good".
- **CMOS 0** = "no difference"; signs are re-coded to the (reference, processed) order before averaging (P.800 Annex E).

Part 2

Objective, signal-based models

Three families of objective models

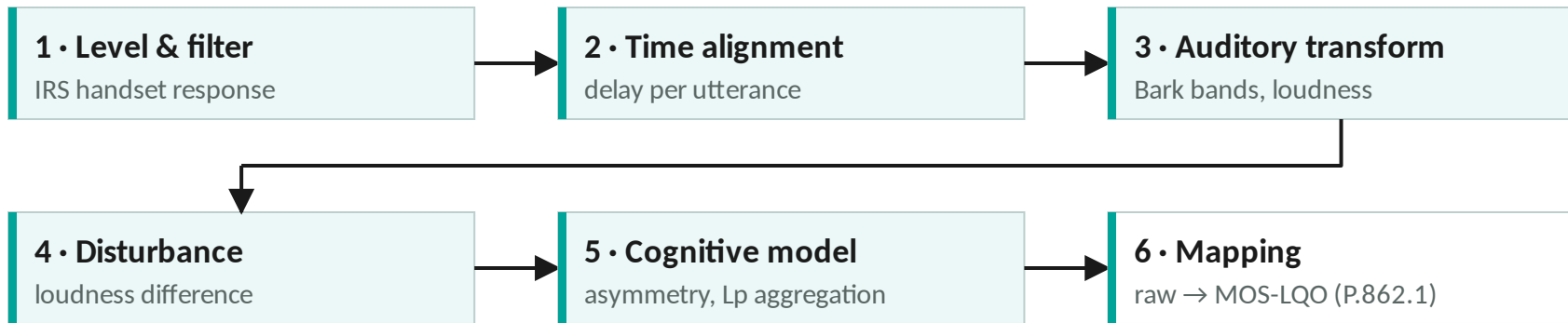


More information in → more accurate, but harder to deploy: a reference signal exists only in a test call.

Intrusive (full-reference) models need a test call with known speech — ideal for laboratories and drive tests. **Non-intrusive** models can monitor live traffic, but must guess what the clean signal was. **Parametric** models need no audio at all and work even at the planning stage.

PESQ — ITU-T P.862 (2001, withdrawn 2024)

INPUT: REFERENCE + DEGRADED SIGNAL



Raw PESQ score -0.5 ... 4.5 · the mapping fits it to subjective ACR results from many listening tests

The core idea: transform both signals into an **internal representation of what the ear perceives** and measure the difference there, not in the waveform. The cognitive model penalises **added** components (noise, artefacts) more than **missing** ones — listeners find them more annoying.

ITU-T deleted the P.862 family in January 2024 in favour of P.863 (POLQA). PESQ remains the most widely published baseline, and its reference code is freely available — which is why Lab 03 still uses it.

From raw PESQ to MOS-LQO — and what PESQ cannot see

The raw score (−0.5 ... 4.5) is mapped to the ACR scale by **P.862.1**:

$$\text{MOS-LQO} = 0.999 + \frac{4}{1 + e^{-1.4945x + 4.6607}}$$

For wideband speech, **P.862.2** uses the same shape with coefficients 1.3669 and 3.8224.

Always report which output you use — raw PESQ and MOS-LQO differ by up to about 0.4 in the middle of the scale.

Outside the scope of P.862 — inaccurate or not intended (Table 2):

- delay in conversation, talker echo — it is a **listening** model
- sidetone, listening level and loudness loss (signals are level-aligned to 79 dB SPL first)
- in-service, non-intrusive measurement

Not validated in the 2001 edition (Table 3): music, codecs below 4 kbit/s, echo cancellers, noise reduction — and **packet loss with PCM codecs and PLC**. Keep that in mind when you score netem-degraded G.711 in Lab 03.

Beyond PESQ — POLQA, ViSQOL and the bandwidth question

Narrowband

G.711, G.729 · P.862 (withdrawn) · P.863 · G.107

Wideband

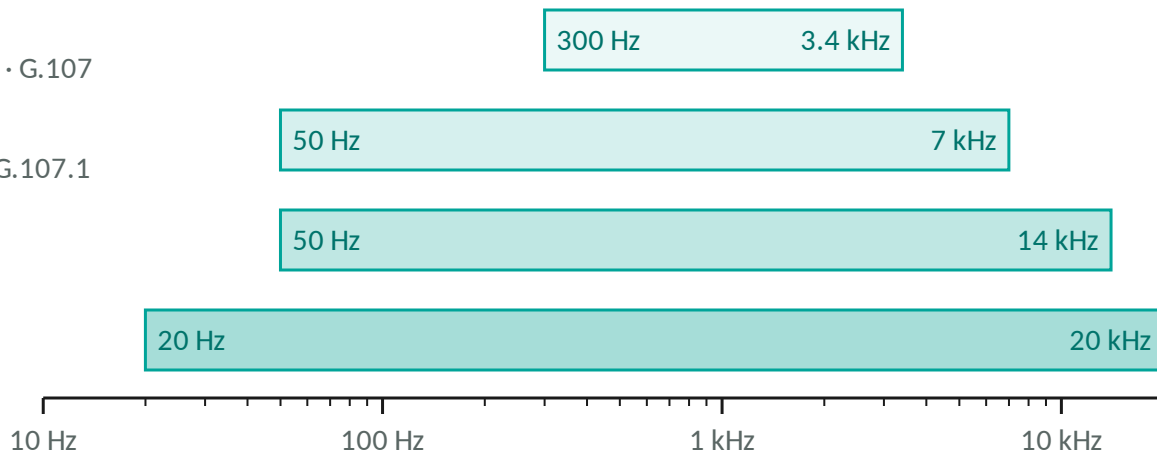
G.722, AMR-WB · P.862.2 (withdrawn) · G.107.1

Super-wideband

EVS, Opus · P.863 · G.107.2

Fullband

EVS, Opus · POLQA P.863 · G.107.2



A model is valid only for the bandwidth it was trained on — never score wideband audio with narrowband PESQ.

POLQA, ITU-T P.863 (2011, current edition 2018) — PESQ's successor and official replacement: narrowband, super-wideband and fullband modes, better handling of time-scaling and modern codecs.

ViSQOL (Google, open source) — compares spectro-temporal similarity of the two signals; speech and audio modes. Used in Lab 03.

Non-intrusive signal-based models

ITU-T P.563 (2004) — single-ended narrowband model

Reconstructs a plausible clean signal from the degraded one using a model of human speech production (vocal tract), then evaluates the distortion classes it detects: noise, temporal clipping, robotisation, unnatural pitch. Accuracy is clearly below the intrusive models.

Learned models — DNSMOS and similar

Deep networks trained directly on large crowdsourced ACR databases (ITU-T P.808 methodology). They predict MOS from the degraded signal alone and are widely used to rank noise suppressors — but they inherit the bias of their training data and are not standardised.

Non-intrusive models are the only signal-based option for **live monitoring**, where the original speech is never available.

Checkpoint — the satellite hop

A call is routed over a geostationary satellite, adding 270 ms of one-way delay. There is no loss and the codec is G.711. You record both ends and run PESQ. What score do you expect, and does it describe the user's experience?

Predict first · discuss with a neighbour · then reveal

PESQ will report nearly clean quality (MOS-LQO above 4) — its time-alignment step removes the constant delay before comparing the signals, exactly as designed. But the delay degrades **conversational** quality: users talk over each other and any residual echo becomes clearly audible. PESQ produces a listening-quality score; the conversational effect has to come from a conversation test or from a model that includes delay — the E-model in Part 3.

Part 3

The E-model — ITU-T G.107

The idea behind the E-model

Developed by ETSI and adopted as ITU-T G.107 (first edition 1998) as a **transmission-planning tool**: predict the quality of a connection that does not exist yet, from the parameters of its components.

Additivity on a psychological scale

Each impairment is expressed as a number on one common scale, the **transmission rating R** (0–100 for narrowband).

Impairments are assumed to be independent, so their contributions are **subtracted** from a best-case value.

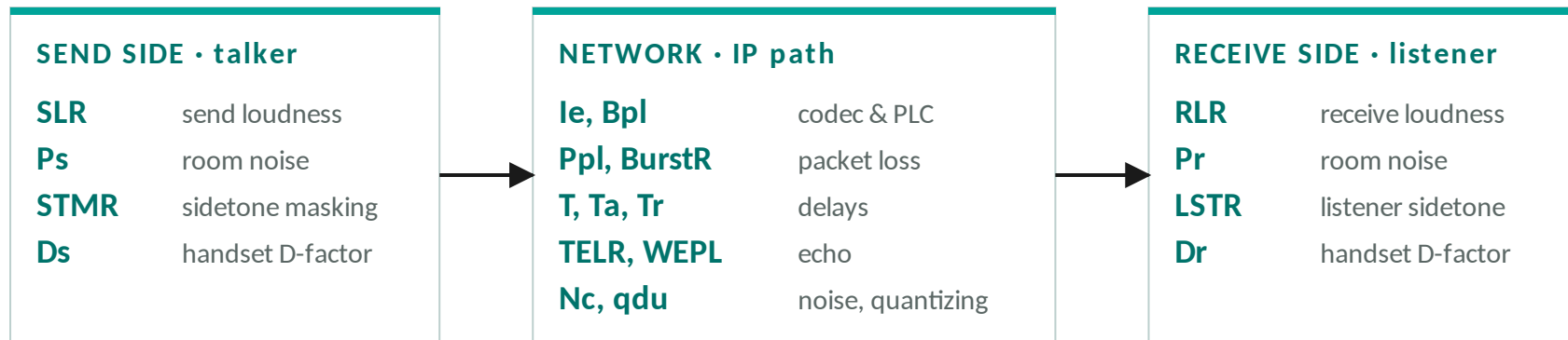
What it is — and is not

The output is an **estimate** (MOS-CQE) for an average user.

G.107 states that such estimates are made "only for transmission planning purposes and not for actual customer opinion prediction".

The reference connection

OLR = SLR + RLR (overall loudness, mouth to ear)



A — advantage (expectation) factor: set by the user's situation, not by the equipment

$$R = Ro - Is - Id - le\text{-}eff + A$$

Around twenty input parameters, each with a **default value** in G.107. For a VoIP study most stay at their defaults; the ones that change are **delay**, the **codec** (le, Bpl), **packet loss** (Ppl, BurstR) and **echo** (TEL).

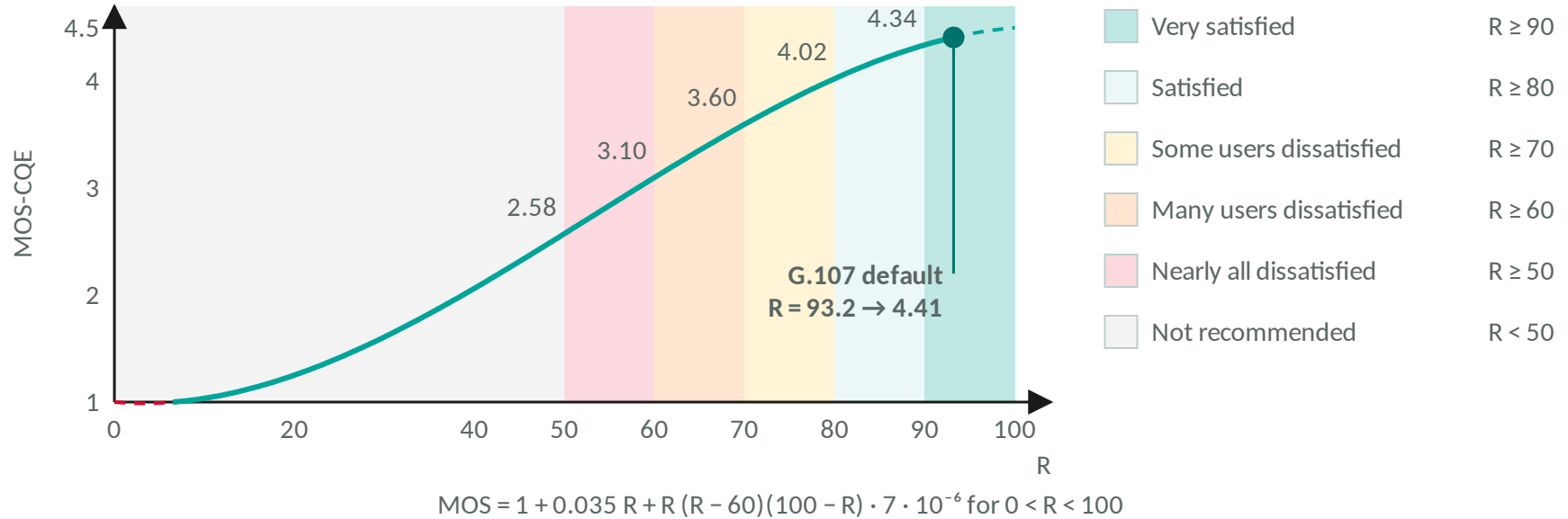
The rating formula

$$R = R_o - I_s - I_d - I_{e\text{-eff}} + A$$

Term	Meaning	Depends on	Default
R_o	basic signal-to-noise ratio	circuit and room noise, loudness	94.77
I_s	simultaneous impairments	loudness, sidetone, quantizing noise	1.41
I_d	delay impairments	one-way delay, echo (TELR, WEPL)	0.15
$I_{e\text{-eff}}$	effective equipment impairment	codec, packet loss, burstiness	0
A	advantage (expectation) factor	the user's situation	0

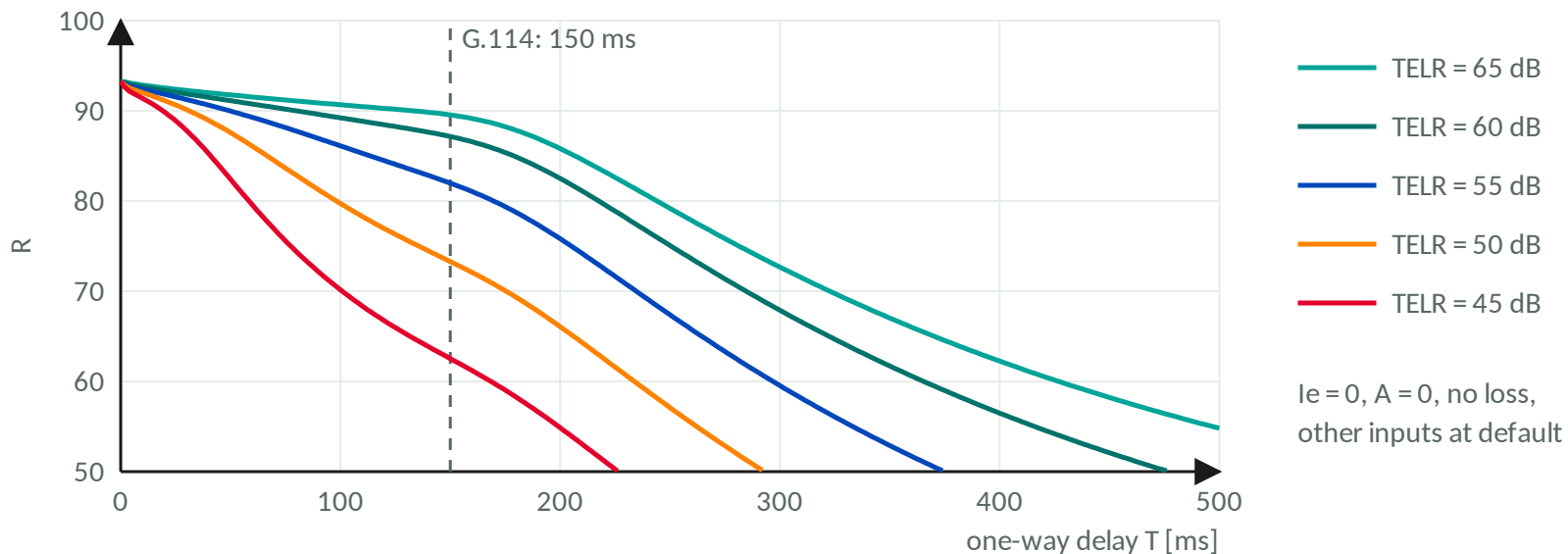
With all defaults: $R = 94.77 - 1.41 - 0.15 - 0 + 0 \approx \mathbf{93.2}$ — the best a narrowband connection can achieve.

From R to MOS and to user satisfaction



For $R \leq 0$, MOS = 1; for $R \geq 100$, MOS = 4.5. The categories are from ITU-T **G.109**; R below 50 is not recommended for any service.

Delay impairment I_d — delay and echo together



Delay alone is tolerable; delay combined with audible echo is what destroys a conversation.

$I_d = I_{dte} + I_{dle} + I_{dd}$: **talker echo**, **listener echo**, and **pure delay** ($I_{dd} = 0$ below 100 ms). A higher TELR means better echo cancellation. The 150 ms guideline from Lecture 03 (ITU-T G.114) is where R starts to fall steeply even with good echo control.

Equipment impairment I_e — the codec

Codec	kbit/s	I_e	Listen
G.711 PCM	64	0	 64
G.726/G.727 ADPCM	40 · 32 · 24 · 16	2 · 7 · 25 · 50	 40  32  24  16
G.728 LD-CELP	16 · 12.8	7 · 20	—
G.729A CS-ACELP	8	10	 8
G.729A + VAD	8	11	 8
G.723.1 MP-MLQ, ACELP	6.3 · 5.3	15 · 19	 6.3

Values from ITU-T G.113, Appendix I, from subjective tests of each codec without transmission errors.

- I_e is **not** proportional to bit rate — LD-CELP at 16 kbit/s beats ADPCM at 24 kbit/s: it models speech, not the waveform.
- Transcoding (G.729 → G.711 → G.729) adds the impairments of every stage.

Reference, 128 kbit/s linear PCM:  original

Speech: Open Speech Repository. G.728 and G.723.1 at 5.3 kbit/s have no freely available encoder.

Effective equipment impairment — adding packet loss

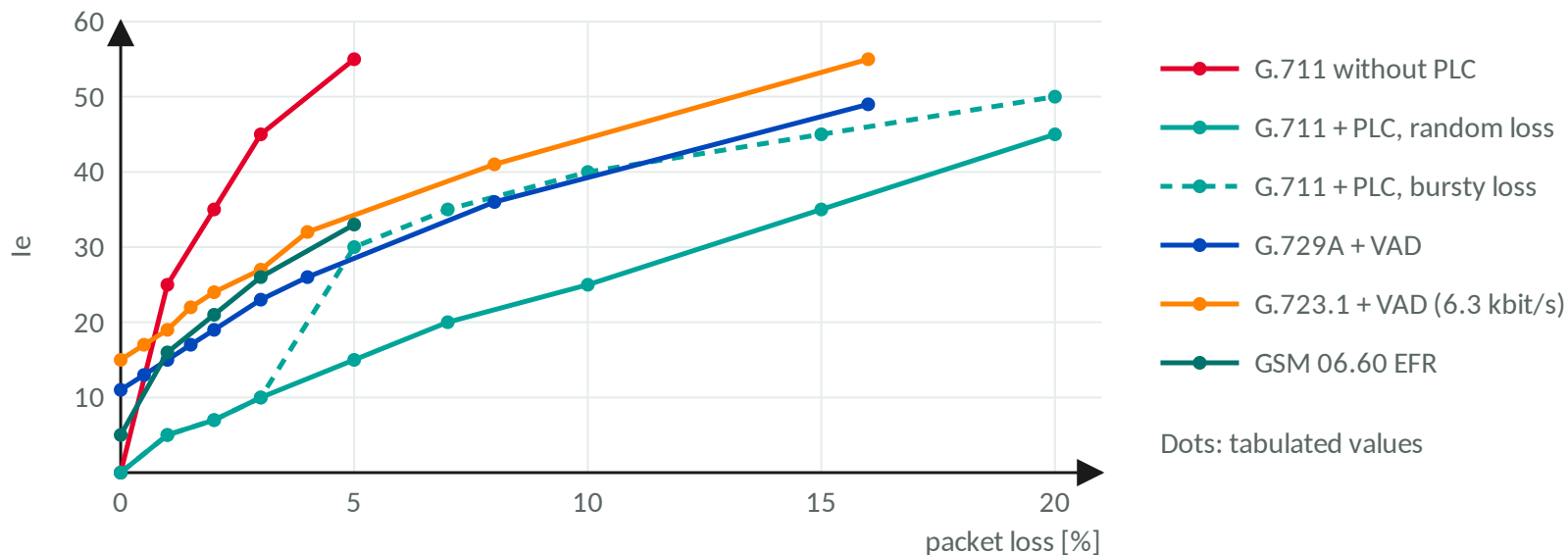
$$I_{e\text{-eff}} = I_e + (95 - I_e) \frac{P_{pl}}{\frac{P_{pl}}{BurstR} + B_{pl}}$$

- P_{pl} — packet-loss probability [%]
- B_{pl} — **packet-loss robustness factor** of the codec and its concealment (PLC), from G.113
- 95 — the ceiling: at total loss, $I_{e\text{-eff}} \rightarrow 95$ and nothing is left

BurstR — burst ratio: the mean length of observed loss bursts divided by the mean length expected for random loss.

- $BurstR = 1$ — random (Bernoulli) loss
- $BurstR > 1$ — bursty loss; for Lecture 04's **simple Gilbert model**
 $BurstR = 1/(p + q)$
- validated only up to $BurstR = 2$, or beyond that for $P_{pl} < 2\%$ (G.107 Annex A)

I_e against packet loss — tabulated values



Provisional planning values of ITU-T G.113 Appendix I (1999) — the data the Bpl formula was later fitted to.

Concealment matters: 1 % loss costs G.711 25 R points without PLC, 5 with it. **So does burstiness:** at 5 % loss, 30 vs 15. The $I_{e\text{-eff}}$ formula was fitted to these values (G.729A, 8 %: table 36, formula 35.9).

B_{pl} and the advantage factor A

Codec	I_e	Packet [ms]	B_{pl}
G.723.1 + VAD	15	30	16.1
G.729A CS-ACELP + VAD	11	20	19.0
GSM EFR	5	20	10.0
G.711, no PLC	0	10	4.3
G.711 with PLC	0	10	25.1

Source: ITU-T G.113 (2024), Appendix I, Table I.4.

Communication system	A (max.)
Conventional wirebound	0
Mobility within a building	5
Mobility in a geographical area or in a vehicle	10
Hard-to-reach locations, e.g. multi-hop satellite	20

Source: ITU-T G.107, provisional values.

A models **user expectation**: people accept lower quality in exchange for convenience or access. It is an upper bound, not a mandatory bonus.

A simplified E-model for VoIP

With terminals and echo control at their defaults:

$$R \approx 93.2 - I_d - I_{e\text{-eff}} + A$$

Cole & Rosenbluth (2001) fit the delay term, for good echo cancellation, with one line and a knee:

$$I_d \approx 0.024 d + 0.11 (d - 177.3) H(d - 177.3)$$

where d is the one-way delay in ms and H the unit step function.

Why it is useful

- Only three measurable inputs — **delay, loss, codec** — all available from RTP/RTCP statistics.
- Easy to compute in a monitoring probe for every call.
- Close to the full model where it matters: at $d = 150$ ms it gives $I_d = 3.6$, the full G.107 computation 3.8.

What it ignores

- noise, loudness, sidetone and echo loss other than the default — and every impairment the default terminal does not have.

Worked example — G.729A over a lossy path

Given: G.729A + VAD ($I_e = 11$, $B_{pl} = 19$), 2 % random loss, one-way delay 150 ms, good echo control, fixed terminal ($A = 0$).

1 • Effective equipment impairment

$$I_{e\text{-eff}} = 11 + 84 \cdot \frac{2}{2 + 19} = 19.0$$

2 • Delay impairment (G.107, TELR = 65 dB): $I_d \approx 3.8$

3 • Rating

$$R = 94.77 - 1.41 - 3.8 - 19.0 + 0 \approx 70.5$$

4 • MOS

$$\begin{aligned} \text{MOS} &= 1 + 0.035 \cdot 70.5 \\ &\quad + 70.5 \cdot 10.5 \cdot 29.5 \cdot 7 \cdot 10^{-6} \\ &= 1 + 2.47 + 0.15 \approx 3.62 \end{aligned}$$

Verdict: *some users dissatisfied*, just above the 70 boundary.

The same path with **G.711 + PLC** ($I_{e\text{-eff}} = 7.0$): $R \approx 82.5$, MOS ≈ 4.12 — *satisfied*. On this path the 56 kbit/s saved by G.729A costs one satisfaction category.

Checkpoint — can the advantage factor rescue a design?

A planner computes $R = 55$ for a VoIP service to a remote research station reached over two satellite hops. They add $A = 20$, obtain $R = 75$, and declare the service acceptable. Is this legitimate?

Predict first · discuss with a neighbour · then reveal

Formally allowed, but it hides the problem. A represents users' willingness to accept worse quality in exchange for access — here plausible, since there is no alternative. But the audio is exactly as bad as $R = 55$ says; A changes the predicted **satisfaction**, not the signal. A defensible report states both values, and first checks whether I_d or $I_{e\text{-eff}}$ can be reduced — echo control or a more robust codec is a real improvement, A is not.

Common mix-ups

MOS-LQO vs. MOS-CQE

PESQ produces a listening score from signals; the E-model estimates a conversational score from parameters. Similar numbers, different meanings.

R vs. MOS

R is additive in impairments; MOS is not. Subtract impairments on the R scale, then convert once.

"PESQ says the call is fine"

PESQ removes delay before scoring and ignores echo and sidetone. It cannot judge a conversation.

I_e vs. I_{e-eff}

I_e is the codec on a perfect network; $I_{e\text{-eff}}$ adds packet loss through B_{pl} and BurstR.

Wrong bandwidth

Narrowband PESQ on wideband audio, or comparing narrowband R (max. 93.2) with wideband R (max. 129), gives meaningless results.

MOS without a confidence interval

A MOS difference smaller than the interval half-width is not a difference.

Summary — three ways to put a number on speech quality

Subjective tests are the reference ACR, DCR and CCR (P.800) produce MOS-LQS or MOS-CQS; always with a confidence interval, always in the context of the test.

Intrusive models compare two signals PESQ, POLQA and ViSQOL model hearing and judgement; accurate for listening quality in their bandwidth, blind to delay and echo.

Non-intrusive models listen to one signal P.563 or learned models such as DNSMOS enable live monitoring at lower accuracy.

The E-model plans from parameters $R = R_o - I_s - I_d - I_{e\text{-eff}} + A$; delay with echo, codec with loss and burstiness, mapped to MOS-CQE and G.109 satisfaction categories.

Exit questions — use the ideas

1. Why is a DCR test more sensitive than an ACR test to a small codec degradation, and what does it cost you?
2. A monitoring system reports MOS 4.3 for every call on a link with 250 ms one-way delay. Which kind of model is it probably using, and what is it missing?
3. Two paths have the same 2 % average loss, one random and one with Gilbert parameters $p = 0.01$, $q = 0.49$. Which gets the higher $I_{e\text{-eff}}$ for G.711 without PLC, and by how much?
4. Why does the E-model's MOS never exceed 4.5, and why is that consistent with Part 1?

Explain the mechanism, not just the name.

Where you use this — Lab 03

Lab 03 builds the whole chain of this lecture on a real VoIP call:

- inject delay, jitter and loss with `tc netem` (Lecture 03)
- extract the audio from the captured RTP stream
- run **ACR and DCR** tests with your classmates → MOS-LQS, DMOS
- score the same recordings with **PESQ and ViSQOL** → MOS-LQO
- compare the rankings — and compute the E-model estimate for the same conditions by hand

Label every number with its P.800.1 name and, for subjective scores, its confidence interval.

```
1  ssh <lab-server>
2  cd qos-03 && uv sync
3  uv run jupyter lab --ip 0.0.0.0
```

References — standards

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Thank you for your attention

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Extras

Wideband E-model and the limits of additivity

Extras — beyond narrowband: G.107.1 and G.107.2

The narrowband E-model tops out at $R \approx 93.2$ — the quality of a good PCM telephone. Wideband and fullband speech are **better than that**, so the scale is extended rather than rescaled:

Model	Bandwidth	$R_{\max}$
G.107	narrowband	≈ 93
G.107.1	wideband	129
G.107.2	fullband	148

A narrowband codec on the extended scale carries an extra impairment for its **missing bandwidth**. This is why a wideband call with some loss can still be rated better than a perfect narrowband call — and why R values must never be compared across the three scales without saying which one is used.

To map an extended-scale R to MOS, G.107.1 first scales it back to the 0–100 range.

Extras — when additivity breaks

The E-model assumes impairments add on the R scale. Subjective tests show systematic deviations (Möller 2000; Raake 2006):

Combined strong impairments

When a low-bit-rate codec meets heavy loss, the measured degradation can be **smaller than the sum** — the listener's quality judgement saturates, while the model keeps subtracting.

Time-varying quality

A burst of loss near the end of a call hurts more than the same burst at the start (**recency effect**). The E-model uses call-level averages and cannot express this; monitoring models that score short windows and pool them over time can.